

CE 606: Constitutive Modeling of Frictional Material

Course Code	CE 606
Course Name	Constitutive Modeling of Frictional Material
L-T-P-C	3-0-0-3
Prerequisites	Geotechnical Engineering I (CE302) or Equivalent/ Mechanics of Solids (ME206) or Equivalent
Intended for	B.Tech. 4th year/ M.S./ M.Tech./ Ph.D.
Distribution	Open Elective
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Mathematical Background:** Overview and importance of constitutive modeling; Preliminaries on tensor; Stress, strain and invariants; Principal stress space: triaxial and octahedral plane; Stiffness tensor; Voigt notation. [6 Lectures]
- **Constitutive Behavior of Geomaterials:** Different laboratory tests and various modeling aspects pertinent to the stress-strain behaviour of geomaterials subjected to shearing and volumetric compression, e.g., void ratio, pressure and stress path dependency, anisotropy and rate/time effects etc. [3 Lectures]
- **Elasticity:** Linear, nonlinear and anisotropic elastic models; Calibration of model parameters. [6 Lectures]
- **Plasticity:** Internal variable, yield criteria, flow rule, plastic potential, Drucker's stability postulate, convexity and normality rules, hardening/ softening, isotropic and kinematic hardening; Different failure criteria for yielding: Tresca, von Mises, Mohr-Coulomb, Drucker-Prager, Hoek and Brown etc. [7 Lectures]
- **Critical State Models:** Critical state concept; Stress-dilatancy theory; Strain hardening and/or work hardening plasticity models for sand: formulation, calibration and implementation to simulate drained and undrained triaxial test. [6 Lectures]
- **Cam-Clay Models:** Cam-Clay models: formulation and calibration ; Implementation of Cam-Clay model to simulate single element tests: consolidation, drained and undrained triaxial test, Application of elasto-plastic models. [10 Lectures]
- **Special topics:** Other advanced constitutive models for frictional materials (e.g., damage plasticity/ viscoplasticity/ unsaturated soil model / disturbed state model etc.) [4 Lectures]

Text Books

1. David Muir Wood, **Soil Behaviour and Critical State Soil Mechanics**, Cambridge University Press, 1991.
2. Alexander Puzrin, **Constitutive Modelling in Geomechanics**, Springer, 2012.

References

1. David Muir Wood, **Geotechnical Modelling**, CRC Press Taylor and Francis Group, 2004.
2. W.F. Chen and E. Mizuno, **Nonlinear analysis in soil mechanics: theory and implementation**, Elsevier Science Ltd, 1990.
3. Hai-Sui Yu, **Plasticity and Geotechnics**, Springer, 2006.
4. W.F. Chen and G.Y. Baladi, **Soil Plasticity: Theory and Implementation**, Elsevier Science Ltd, 1985.
5. C.S. Desai, **Mechanics of Materials and Interfaces: The Disturbed State Concept**, CRC Press, 2000.
6. N.S. Ottosen and M. Ristinmaa, **The Mechanics of Constitutive Modeling**, Elsevier Science, 2005.
7. J.C. Simo and T.J.R. Hughes, **Computational Inelasticity**, Springer, 2000.
8. E.A. de Souza Neto, D. Peric and D.R.J. Owen, **Computational Methods for Plasticity: Theory and Applications**, Wiley, 2008.
9. Allan F. Bower, **Applied Mechanics of Solids**, CRC Press, 2009.